

ASCII MASTER FLOW – PANEL 1/12: Laurentian climate world distribution

NE NORTH AMERICA -> eastern Canada, Maritime Provinces, New England, Newfoundland

EAST ASIA -> eastern Siberia, North China, Manchuria, Korea, northern Japan

SOUTHERN HEMISPHERE -> absent: continents too narrow at relevant latitudes

INTERMEDIATE -> between British maritime (west) and Siberian continental (east)

MUST REMEMBER: Cool temperate eastern-margin Laurentian climate combines continental air masses, maritime moisture, warm-current influence and frontal/cyclonic precipitation, yielding warm summers, cold winters and year-round precipitation in northeastern North America and northeast Asia.

ASCII MASTER FLOW – PANEL 2/12: Seasonal rainfall mechanism

WINTER -> continental Westerlies blow dry frigid air from interior
SUMMER -> easterly ocean winds bring moisture-laden rainfall
RESULT -> distinct summer maximum: two-thirds of 30-60 inches total
CONTRAST -> reverse of British type's autumn-winter rain pattern

ASCII MASTER FLOW – PANEL 3/12: Current convergence and fishing

OYASHIO (cold) + KUROSHIO (warm) -> convergence off northern Japan

LABRADOR (cold) + ATLANTIC (warm) -> convergence over Grand Banks shelf

NUTRIENT MIXING -> plankton bloom on broad shallow shelves

RESULT -> world-historic fishing grounds plus persistent fog

ASCII MASTER FLOW – PANEL 4/12: Fog as dual phenomenon

FOG AS HAZARD -> poor visibility, navigational danger, shipping risk

FOG AS INDICATOR -> biological marker of nutrient-rich mixing waters

MECHANISM -> warm moist air chilled from below by cold current contact

EXAM USE -> same process creates hazard and resource simultaneously

ASCII MASTER FLOW – PANEL 5/12: Mixed forest transition strip

BRITISH MARGIN -> deciduous broadleaf forest (mild, wet winters)

LAURENTIAN ZONE -> mixed coniferous-deciduous transition forest

SIBERIAN INTERIOR -> evergreen coniferous taiga (cold, dry winters)

GRADIENT -> maritime to continental controls the forest composition

ASCII MASTER FLOW – PANEL 6/12: Fishery governance lesson

PHYSICAL BASE -> current convergence + shallow shelf + nutrient mixing

INITIAL OUTCOME -> one of the world's most productive fisheries

OVEREXPLOITATION -> Grand Banks cod stocks collapsed despite same oceanography

CONCLUSION -> governance, not geography, determines sustainability

ASCII MASTER FLOW – PANEL 7/12: Eastern Himalaya analogue map

DARJEELING-SIKKIM -> cool humid eastern-facing temperate belt

ARUNACHAL PRADESH -> warm perhumid ecoregion, growing period > 210 days

MONSOON RAINFALL -> summer maximum parallels Laurentian pattern

HILL FOG -> persistent mist mirrors Newfoundland-Japan fog zones

ASCII MASTER FLOW – PANEL 8/12: Teesta basin profile

STEEP RELIEF -> rapid runoff, high sediment load

EXTREME MONSOON RAIN -> triggers landslides and GLOF events

CORRIDOR CONCENTRATION -> NH-10, hydropower on narrow valley floor

2023 SOUTH LHONAK GLOF -> illustrates combined relief-rain-infrastructure risk

ASCII MASTER FLOW – PANEL 9/12: Darjeeling tea economy

DARJEELING HILLS -> low-yield, premium, GI-protected orthodox tea

BRAHMAPUTRA VALLEY -> bulk-tea belt, high-volume production

ELEVATION -> estates at 900-1800 m; too cold above, too warm below

DISTINCTION -> hill quality versus valley quantity mirrors altitude control

ASCII MASTER FLOW – PANEL 10/12: British-Laurentian comparison

RAINFALL -> British: autumn-winter max | Laurentian: summer max

WINTER -> British: mild, wet (maritime) | Laurentian: cold, dry (continental)

ECONOMY -> British: land-based agriculture | Laurentian: sea-based fishing

FOREST -> British: deciduous broadleaf | Laurentian: mixed transitional

CLOSE DISTINCTION: Laurentian differs from warm China type by colder winters and from Siberian interiors by stronger maritime precipitation; warm and cold currents modify coastal gradients but do not alone create the whole climate.

ASCII MASTER FLOW – PANEL 11/12: Polar-vortex hypothesis

STRATOSPHERIC DISRUPTION -> weakens polar vortex containment

COLD ARCTIC AIR -> spills southward into mid-latitudes

LAURENTIAN REGIONS -> experience severe cold outbreaks

STATUS -> actively debated link to Arctic sea-ice loss; not settled science

EVIDENCE LIMIT: Current-air-mass causation must be expressed as interaction with latitude, westerlies and continentality; Eastern Himalayan temperate zones are topographic analogues rather than Laurentian climatic replicas.

ASCII MASTER FLOW – PANEL 12/12: Laurentian answer spine

DEFINE -> intermediate type between British maritime and Siberian continental

LOCATE -> NE North America and East Asia only; absent in Southern Hemisphere

EXPLAIN -> cold dry winters, summer rain maximum, current convergence, fog-fishery

QUALIFY -> India Eastern Himalaya analogue; transparent zero-direct-PYQ boundary